

Public Data Insights Dashboard

WEEK-04

Objective:

The aim of this project is to analyze the cleaned Netflix dataset using different data visualization techniques. These visualizations help identify trends, compare different categories, and uncover patterns within the data. Presenting the information graphically makes it easier to understand the dataset and draw meaningful insights for better decision-making. In this project, we use a Streamlit app to create an interactive dashboard that makes it easier to understand data trends and support better decision-making.

Dataset Overview:

The analysis was performed on the cleaned Netflix dataset containing information about movies and TV shows available on Netflix.

Data Validation

Before creating visualizations, the cleaned dataset was verified to ensure data quality.

The following checks were performed:

- Missing value inspection
- Duplicate record detection
- Column verification
- Dataset preview

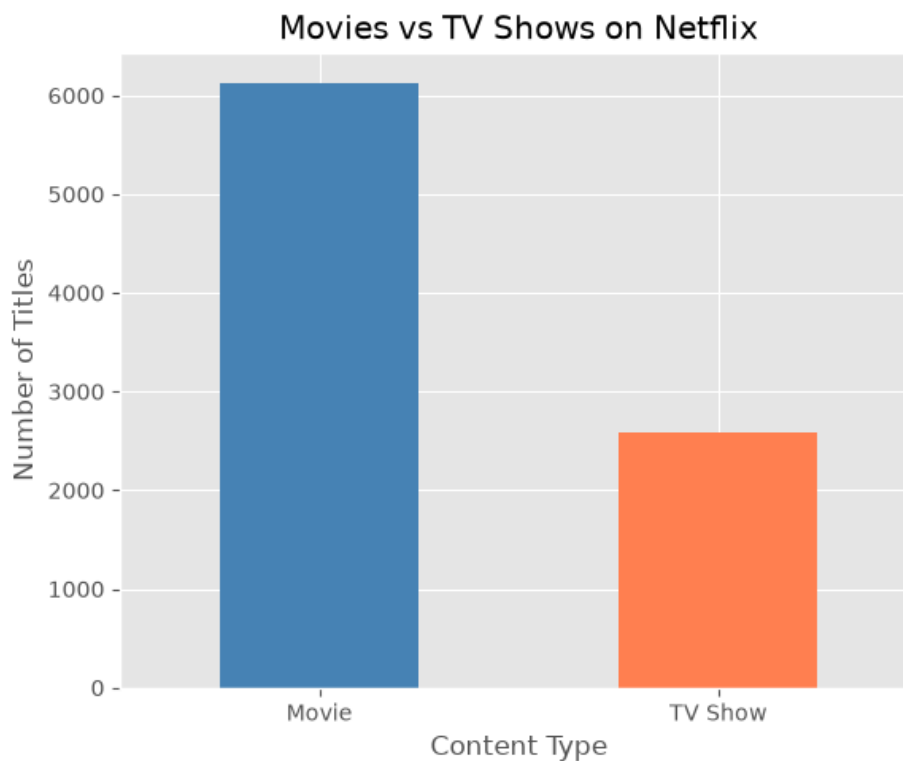
Visualizations Performed:

Now we are going to create some charts to understand the dataset and what insights it is giving.

Bar Chart – Movies vs TV Shows

To compare the number of Movies and TV Shows available on Netflix.

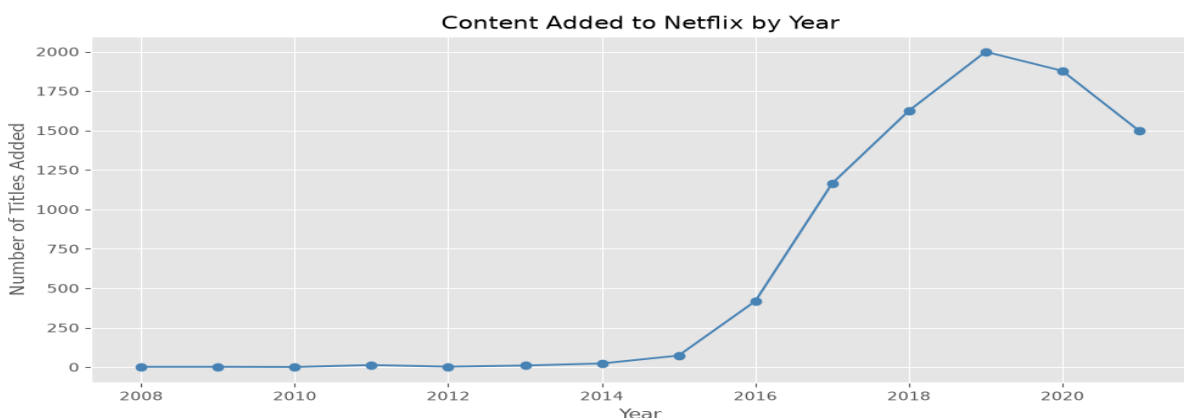
To analyze how many titles were added to Netflix each year.



The chart clearly shows that Movies dominate Netflix's catalog, accounting for approximately 70% of all available content, while TV Shows represent a much smaller portion.

Line Chart – Content Added Over Time

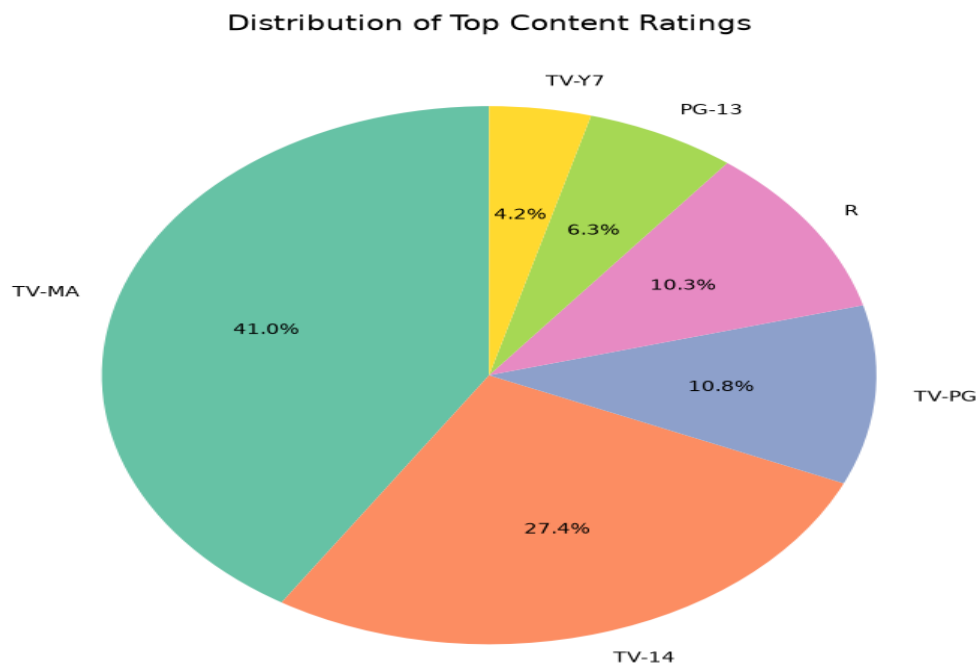
To analyze how many titles were added to Netflix each year.



The number of titles added increased steadily over the years and reached its highest point around 2019. After this peak, the number of additions declined slightly during 2020 and 2021.

Pie Chart – Top Content Ratings

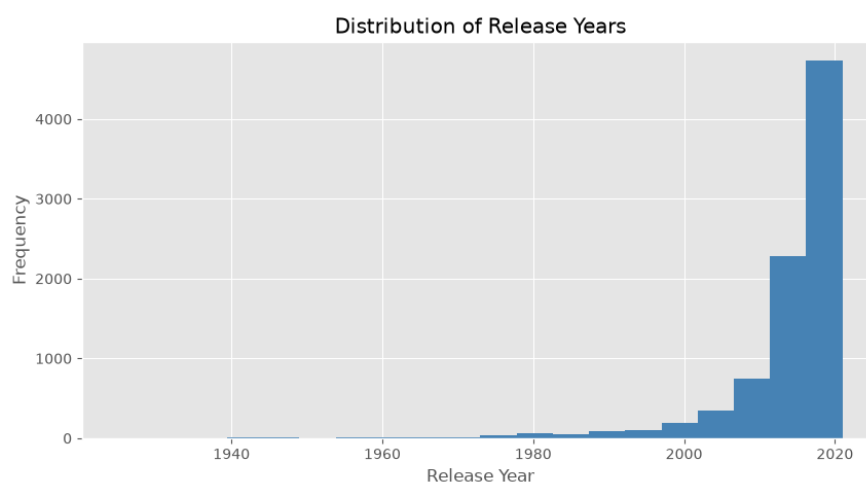
To examine the distribution of the most common content ratings.



TV-MA occupies the largest portion of the chart, followed by other ratings such as TV-14 and TV-PG.

Histogram – Release Year Distribution

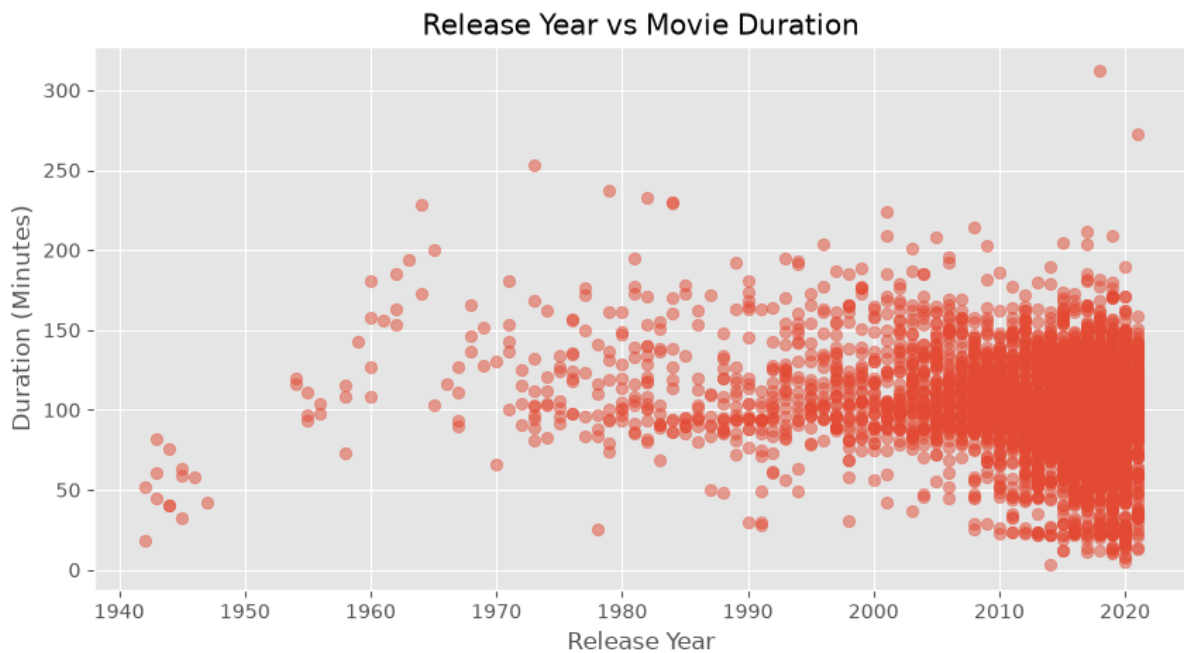
To observe how Netflix content is distributed across different release years.



Most titles were released after 2013, while comparatively fewer titles belong to earlier years.

Scatter Plot – Release Year vs Movie Duration

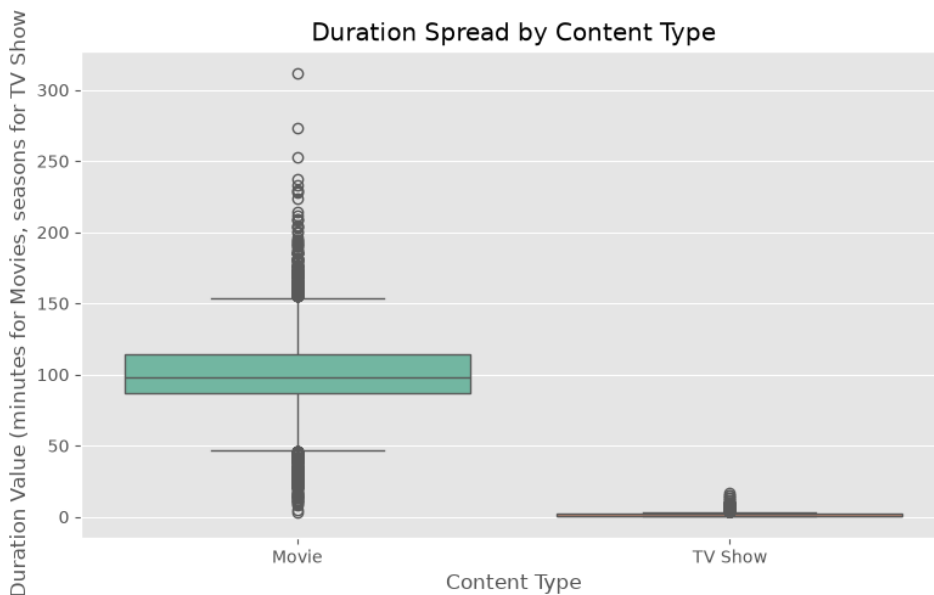
To investigate whether movie duration has changed over time.



The scatter plot shows a slight downward trend, indicating that newer movies tend to be somewhat shorter.

Box Plot – Duration Comparison

To compare the distribution of movie durations and TV show seasons.



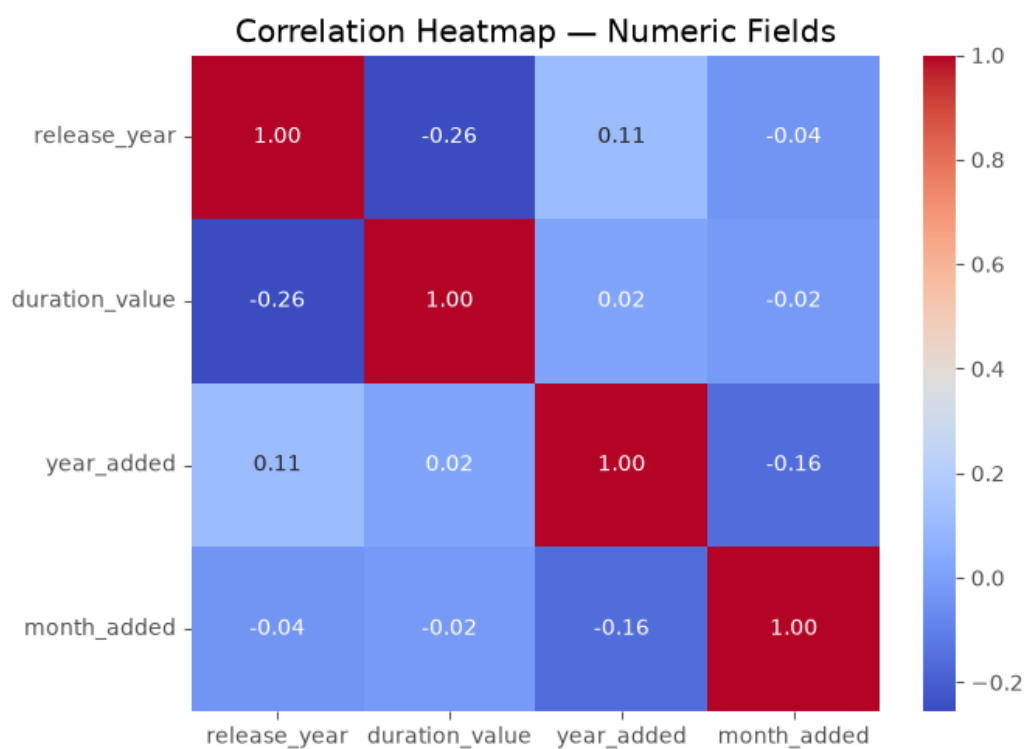
Movies generally range between approximately 87 and 114 minutes, while most TV Shows consist of only one or two seasons, with a few long-running outliers.

Heatmap – Correlation Analysis

To identify relationships among numerical variables.

Variables analyzed:

- Release Year
- Duration Value
- Year Added
- Month Added

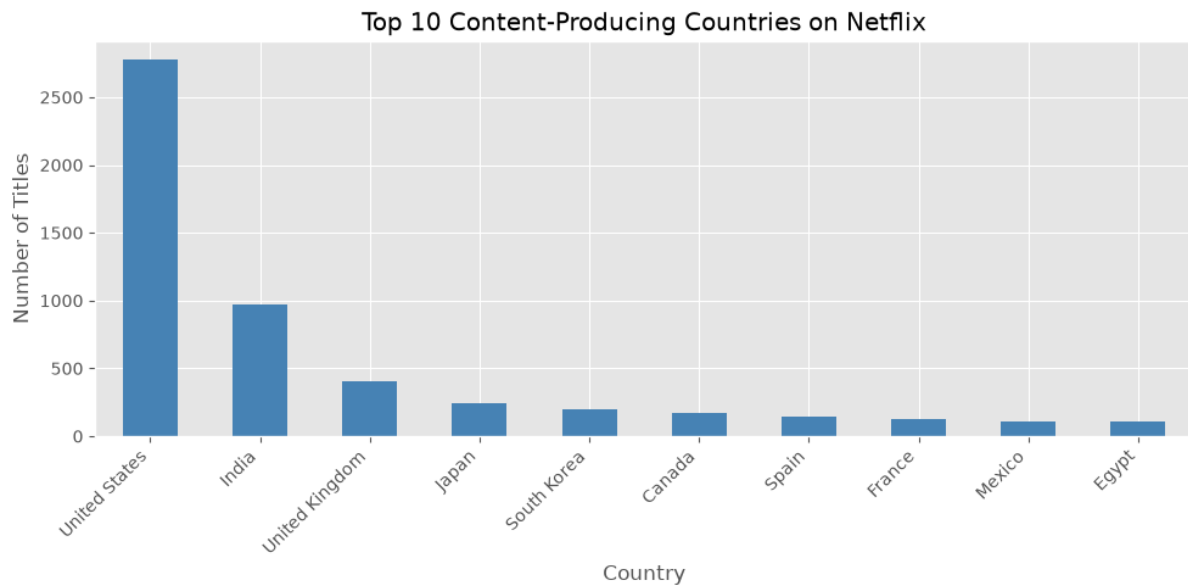


The heatmap shows mostly weak correlations between variables.

The strongest relationship appears between release year and duration, although it remains relatively weak.

Category Comparison – Top Producing Countries

To compare the countries contributing the highest number of Netflix titles.



The United States contributes the largest number of titles, followed by India and several other countries.

Streamlit-app

The Streamlit dashboard was developed to provide an interactive platform for exploring and analyzing the cleaned Netflix dataset. It allows users to visualize the data through various charts and graphs while applying filters based on content type, release year, and country. The dashboard dynamically updates all visualizations according to the selected filters, making data exploration simple and interactive.

Dashboard Features

Interactive Filters: To support dynamic exploration of the dataset, three filtering options are provided:

- Content Type
- Release Year Range
- Country

Dataset Overview

A summary section is displayed at the beginning of the dashboard to provide an overview of the filtered dataset. It includes:

- Total Titles
- Number of Movies
- Number of TV Shows
- Countries Represented

Data Visualizations

The dashboard contains multiple interactive visualizations that present different aspects of the Netflix dataset, including:

- Top 10 Countries by Number of Titles
- Titles Released Over Time
- Distribution of Content Ratings
- Distribution of Release Years
- Release Year vs. Movie Duration
- Duration Spread by Content Type
- Correlation Heatmap
- Top Genres

Insights

The Insights section highlights the major observations obtained from the dashboard analysis. It explains that Netflix's catalog is dominated by movies rather than TV shows, most content targets mature audiences, and the majority of titles are produced by the United States and India. It also emphasizes that the platform contains fewer family-oriented titles and limited content from many other countries.

Recommendations

The Recommendations section presents prioritized suggestions (**High**, **Medium**, and **Low** priority) based on the analysis. It recommends expanding the TV show catalog to improve viewer engagement, increasing family- and children-oriented content to reach a wider audience, and investing in content production from underrepresented countries to improve global content diversity.

Conclusion

This dashboard successfully transformed the cleaned Netflix dataset into meaningful visual representations that simplify data interpretation. Each visualization highlighted a different aspect of the dataset, making it easier to identify trends, compare categories, and understand the characteristics of Netflix's content library.

The analysis revealed important insights into Netflix's content distribution, release trends, ratings, genres, duration, and country-wise production. By presenting these findings through an interactive Streamlit dashboard, users can easily explore the dataset, apply filters, and gain a deeper understanding of the data. Overall, the dashboard demonstrates how data visualization can convert raw data into valuable insights that support informed decision-making and help identify opportunities for improving Netflix's content strategy.